
a
Atrial measurement as a structured, beat-resolved problem
Frame states
Ordered / null
on ≤ peak ≤ off
Continuous spans
R
Unique beat slot
Whole-patient input · V1 excerpt shown
b
Ordered/null policy improves precision
II
V1
Conservative precision (%)
Small → large: continuous → R → ordered/null
Reference-P recall (%)
c
Precision gains across both seeds
Seed1
Seed2
Ordered/null − R precision (pp)
+4.90
+6.91
+5.19
+5.84
Lead II
Lead V1
d
Observation-aware learning under controlled missingness
Candidate features
Train:20% / 40% M masks
Stress test:0–80%
D1 explicit morphology
D2 confidence weighting
D3 weighting + mask intervention
V / N are retained as observation-process information; unknown morphology is not imputed as physiological no-P.
e
Smaller perturbation of AF probabilities
D0
D1
D2
D3
Mean |Δp(AF)| (×10⁻³)
Artificial M masking (%)
f
Probability error under artificial masking
D0: no-M comparator
Brier score (×10⁻³)
Artificial M masking (%)
Component-level engineering evidence; artificial masking is not a physiological intervention or independent validation.